PERFORMANCE EVALUATION OF HVOF-SPRAYED STELLITE 6 COATING ON TUNGSTEN–COPPER ALLOY FOR HIGH-SPEED PANTOGRAPH APPLICATIONS

A PROJECT WORK PHASE – I REPORT

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BONAFIDE CERTIFICATE

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DECLARATION

We hereby affirm that the project report titled **"PERFORMANCE EVALUATION OF HVOF-SPRAYED STELLITE 6 COATING ON TUNGSTEN–COPPER ALLOY FOR HIGH-SPEED PANTOGRAPH APPLICATIONS"** being submitted in partial fulfilment of the requirements for the award of the **BACHELOR OF ENGINEERING IN MECHANICAL ENGINEERING** is original work carried out by us. It has not formed part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

The advancement of high-speed rail is limited by the performance of pantograph contact strips, where traditional materials face a critical trade-off between electrical conductivity and wear resistance. This project develops a novel surface-engineered solution by applying a Stellite 6 coating via the High-Velocity Oxygen Fuel (HVOF) process onto a tungsten-copper (W-Cu 80/20) substrate. The W-Cu base provides excellent structural integrity and current-carrying capacity, while the Stellite 6 layer is designed to offer superior protection against mechanical wear, arc erosion, and thermal fatigue. This Phase-I report details the successful completion of the foundational work. A certified W-Cu alloy was procured, and a set of test specimens with specific geometries was precision-fabricated using Wire Electrical Discharge Machining (EDM) to ensure dimensional accuracy and surface integrity without inducing damage. All specimens underwent rigorous quality verification, confirming their readiness for the subsequent coating phase. This meticulous preparation establishes a robust platform for Phase-II, which will involve HVOF coating deposition and comprehensive performance evaluation. The proposed composite system is expected to significantly enhance service life and operational reliability, addressing a key technological bottleneck in modern railway systems. The study thus lays the groundwork for developing next-generation high-performance contact materials capable of sustaining demanding operational environments, contributing to improved efficiency, reduced maintenance, and long-term sustainability in high-speed rail applications. Furthermore, the research emphasizes material optimization through microstructural refinement, coating adhesion enhancement, and improved surface finish, ensuring reliable current collection under fluctuating loads. The outcomes are anticipated to promote innovation in surface coating technologies for advanced rail and power transmission systems.

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ABBREVIATIONS

Abbreviation	Full Form
A	Ampere
CNC	Computer Numerical Control
EDM	Electrical Discharge Machining
EDX	Energy Dispersive X-ray Spectroscopy
HSR	High-Speed Rail
HV	Vickers Hardness
HVOF	High-Velocity Oxygen Fuel
IACS	International Annealed Copper Standard
LSA	Laser Surfaced Alloying
OM	Optical Microscopy
PCS	Pantograph-Catenary System
SEM	Scanning Electron Microscopy

SYMBOLS AND GREEK LETTERS

Symbol	Description
°C	Degree Celsius
Cr_7C_3, $Cr_{23}C_6$	Chromium Carbide Phases
Hz	Hertz
mm	Millimetre
N	Newton
W/m·K	Watt per Meter-Kelvin (Thermal Conductivity)

NOMENCLATURE

Term	Definition / Context
Adhesive Wear	Wear due to localized bonding between contacting solid surfaces.
Arc Erosion	Material loss from a surface caused by an electric arc.
Catenary	The overhead wire system from which the pantograph collects electricity.
Contact Strip	The consumable component on the pantograph that slides against the catenary wire.
Dielectric Fluid	A non-conductive fluid used in EDM to control sparking and flush debris.
Functionally Graded Material	A material with properties that gradually change with position.
Microstructure	The small-scale structure of a material, as revealed by a microscope.
Monolithic Material	A material with a uniform composition and structure throughout.
Pantograph	The articulated assembly on a train roof that collects current from the catenary.
Porosity	The measure of void spaces in a material.
Powder Metallurgy	A process of forming metal parts by heating compacted powders.
Pseudo-Alloy	A composite material made of immiscible components (e.g., W-Cu).
Stellite 6	A cobalt-based superalloy known for high wear and

	corrosion resistance.
Substrate	The base material upon which a coating is applied.
Surface Engineering	The application of a coating or treatment to a surface to enhance its properties.
Thermal Fatigue	Cracking or damage caused by repeated cycles of heating and cooling.
Thermal Spray	A coating process where melted materials are sprayed onto a surface.
Tribology	The science and engineering of interacting surfaces in relative motion.
Wire EDM	A precision machining process using a wire electrode to cut conductive materials.

CHAPTER 1: INTRODUCTION

1.1 General

The evolution of global transportation infrastructure has unequivocally positioned railway systems as a cornerstone of modern sustainable mobility. They offer an unparalleled combination of efficiency, high passenger/freight capacity, safety, and a reduced environmental footprint compared to other mass transit options. This prominence is particularly magnified in the rapid global development of high-speed rail (HSR) networks, which represent one of the most significant engineering achievements in modern transit history. The technological metamorphosis from rudimentary steam locomotives to sleek, aerodynamic electric trains capable of sustained operation at speeds exceeding 300–350 kilometers per hour reflects over a century of relentless innovation across multiple disciplines, including mechanical design, aerodynamics, advanced materials science, and electrical engineering.

At the heart of any electric train lies the critical need for a continuous and reliable power supply. This is the singular function of the pantograph-catenary system (PCS), a dynamic and complex interface where a moving train collects electricity from a stationary overhead line. The pantograph, a mechanically articulated assembly mounted on the train roof, is tasked with maintaining uninterrupted physical and electrical contact with the catenary wire under all operational conditions. The most critical component of this assembly is the contact strip, a consumable element that bears the brunt of the extreme operational environment. It must facilitate the steady transfer of thousands of amperes of electrical current while simultaneously sliding at high velocity against the hard copper-alloy wire, enduring intense mechanical friction, transient electrical arcing during momentary loss of contact, and substantial thermal loading from both Joule heating and frictional work. The performance and longevity of this humble strip are, therefore, directly linked to the operational efficiency, safety, and maintenance costs of the entire railway system. As operators worldwide push the boundaries of speed, frequency, and payload, the limitations of traditional contact strip materials have become a critical bottleneck, necessitating a paradigm shift in material design and engineering.

1.2 The Pantograph–Catenary System

The pantograph-catenary system is a masterpiece of precision engineering, designed to solve the deceptively simple yet profoundly challenging task of delivering megawatts of electrical power to a vehicle moving at extremely high speeds. Its operational success hinges on a finely tuned balance between mechanical dynamics, electrical conductivity, material science, and aerodynamic stability.

The system comprises two primary elements: the catenary, a fixed overhead network of suspended conductors, and the pantograph, the spring-loaded mechanism on the train that uplifts to make contact. The catenary is not a simple straight wire; it is a carefully tensioned assembly suspended in a specific zig-zag pattern along the track's centerline. This design is intentional, ensuring that the contact strip wears evenly across its width, preventing the formation of localized grooves that would otherwise catastrophically shorten its life and damage the contact wire. The contact wire itself is typically manufactured from high-conductivity copper or copper-cadmium alloy, engineered to carry high electrical power—commonly 25 kV alternating current at 50 Hz in modern systems or 1.5–3 kV direct current in older or metro systems.

The pantograph is far from a passive component; it is a lightweight, often aerodynamically profiled frame, fabricated from aluminium or carbon-fiber composites to minimize inertia and air resistance. It is pneumatically or spring-assisted to raise and lower, and at its apex resides the critical contact strip. A paramount operational parameter is the contact force—the upward pressure the pantograph exerts on the wire. This force must be maintained within a narrow optimal window. Insufficient force leads to contact loss, causing micro-arcs that erode surfaces and generate electromagnetic interference. Excessive force accelerates mechanical wear on both the strip and the expensive catenary wire, and can induce dangerous aerodynamic oscillations. At speeds beyond 300 km/h, the dynamics become intensely complex, with aerodynamic lift, vibrations from track irregularities, and pressure waves from tunnels causing the pantograph to bounce, leading to frequent arc-induced damage. This combination of extreme stresses makes the PCS interface one of the most demanding tribological and electrical contact environments in all of engineering, as illustrated in **Figure 1.1.**

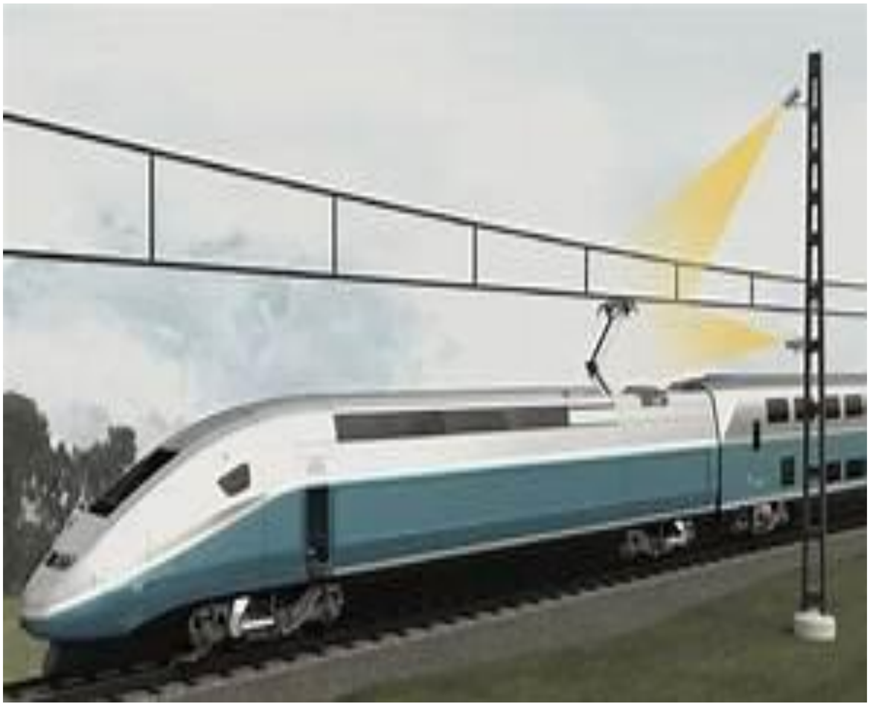

Figure 1.1: High-Speed Train with Pantograph System

1.3 Challenges in Pantograph Contact Strips

The operational lifespan of a pantograph contact strip is dictated by its ability to withstand a synergistic and severe combination of degradation mechanisms that operate simultaneously. The most fundamental challenge is wear, an inevitable consequence of sliding contact over millions of meters. However, this wear is not a singular phenomenon but a complex interplay of multiple modes. Mechanical abrasion occurs when hard contaminant particles, oxide debris, or wear fragments become trapped at the interface, acting as a grinding medium. Adhesive wear takes place when, under high local pressure and temperature, microscopic surface asperities on the strip and wire momentarily weld together and are subsequently torn apart as sliding continues, leading to material transfer and surface roughening.

Arc erosion is the most destructive mechanism. When the contact is momentarily lost due to vibration or wire irregularities, the electrical circuit attempts to bridge the gap by forming an electric arc. These arcs are localized plasma discharges reaching temperatures exceeding 10,000°C, capable of instantaneously melting and vaporizing microscopic volumes of the contact surface. Each arcing event creates pits, cracks, and a roughened morphology that predisposes the surface to further arcing, creating a vicious cycle of degradation. Compounding these mechanical and electrical assaults is intense thermal stress. The combined effect of frictional heating from sliding and resistive (Joule) heating from the high current can elevate localized surface temperatures to several hundred degrees Celsius. At these temperatures, materials soften, oxidize rapidly, and undergo microstructural changes that accelerate wear. Furthermore, the cyclic nature of this heating and cooling can induce thermal fatigue, leading to crack initiation and propagation. The operating environment adds another layer of complexity, exposing the strip to rain, humidity, industrial pollutants, and salt, which can promote corrosion and alter friction characteristics.

The core material science challenge lies in the intrinsic property trade-off: high-conductivity materials like copper are soft and wear rapidly, while hard, wear-resistant materials like ceramics or certain composites are typically poor electrical conductors. This fundamental conflict has long limited the performance of monolithic materials, demanding an innovative engineering solution that decouples the surface requirements from the bulk material needs.

1.4 Proposed Surface Engineering Solution

Surface engineering emerges as a powerful and technologically sophisticated strategy to circumvent the inherent limitations of monolithic materials. Instead of searching for a single, non-existent "ideal" material that excels in all required properties, this approach enables the functional grading of a component. It strategically decouples the requirements, allowing a substrate to be selected for its bulk properties—such as structural strength, high electrical and thermal conductivity, and toughness—while a specialized surface layer is engineered to provide a completely different set of properties, such as extreme hardness, wear resistance, and thermal stability.

For the pantograph contact strip application, this paradigm is perfectly suited. This research proposes the development of a novel composite system where a tungsten-copper (W-Cu 80/20) substrate is coated with a Stellite 6 layer applied via the High-Velocity Oxygen Fuel (HVOF) thermal spray process.

The W-Cu substrate is a well-established pseudo-alloy for demanding electrical contacts. Manufactured via powder metallurgy, it consists of a hard, refractory tungsten skeleton infiltrated with a continuous network of highly conductive copper. This unique microstructure provides an excellent compromise, offering substantial mechanical strength, exceptional resistance to arc erosion, and acceptable electrical conductivity (typically 20-25% IACS), making it a far more robust foundation than pure copper or bronze.

The coating material, Stellite 6, is a cobalt-based superalloy famous for its outstanding performance in severe tribological and high-temperature environments. Its microstructure comprises a tough, ductile cobalt-chromium matrix reinforced with a fine, hard dispersion of chromium carbides. This combination grants it remarkable resistance to abrasion, adhesion, galling, and oxidation, even at elevated temperatures.

The HVOF process is critical to the success of this union. It propels Stellite 6 powder particles at supersonic speeds, impacting the substrate to form a dense, well-bonded coating with very low porosity and minimal oxidative degradation. The central hypothesis is that this thin, durable Stellite 6 coating will act as a protective shield, drastically reducing wear and arc erosion, while the W-Cu substrate ensures efficient current collection and structural integrity. This synergistic combination is expected to yield a next-generation contact strip capable of meeting the rigorous demands of modern high-speed rail

CHAPTER 2: LITERATURE REVIEW

2.1 General

This chapter presents a comprehensive review of existing literature and industrial practices related to the development of advanced pantograph contact strips. It traces the evolution of contact materials, examines the properties of the selected substrate and coating materials, and reviews the deposition technologies used in their fabrication. The objective is to establish the theoretical and practical foundation for this research while identifying key knowledge gaps that justify the present investigation. The discussion begins with the historical progression of pantograph strip materials, highlighting the trade-offs between electrical conductivity, wear resistance, and mechanical integrity that have guided material innovation. Early materials such as copper and carbon composites offered excellent conductivity but suffered from poor wear and arc erosion resistance. These limitations prompted the exploration of metal-ceramic and metal-metal composites, which improved performance but still faced degradation under severe electrical and thermal conditions. As a result, recent trends emphasize surface engineering approaches—particularly thermal spray coatings—to enhance the durability of conductive substrates.

Among potential substrate materials, tungsten-copper (W–Cu) composites have gained prominence due to their unique combination of thermal conductivity, mechanical strength, and resistance to arc erosion. However, their surface wear performance requires further improvement for high-speed applications, making them ideal candidates for protective coatings. In parallel, the High-Velocity Oxygen Fuel (HVOF) thermal spray process has emerged as a superior technique for depositing dense, adherent, and low-oxide coatings. Within this context, Stellite

6—a cobalt-based superalloy—has demonstrated exceptional wear, oxidation, and corrosion resistance while maintaining toughness at elevated temperatures. The integration of an HVOF-sprayed Stellite 6 coating on a W–Cu substrate represents an innovative approach to achieving both electrical efficiency and surface durability. This chapter, therefore, synthesizes prior research to justify this material pairing and outlines the technological gap that this study seeks to address for next-generation high-speed pantograph systems.

2.2 Introduction to Pantograph-Catenary Tribology and Material Requirements

The pantograph–catenary interface is vital for electric rail systems, ensuring continuous current collection under mechanical vibration, aerodynamic forces, and environmental exposure. The efficiency and durability of this interface depend on the material properties of both the overhead contact wire and the pantograph strip. Conventional materials such as pure carbon, copper-impregnated carbon, and metal composites each have limitations—carbon offers good arc resistance but low conductivity and high wear, while metals provide high conductivity but suffer from arc erosion and adhesion at high speeds. With the growing demands of high-speed rail, these materials no longer meet performance requirements.

To address these challenges, advanced composite structures are being developed, combining a conductive substrate with a protective coating to achieve superior wear, oxidation, and arc resistance. Among the promising options, a Tungsten–Copper (W–Cu) alloy substrate coated with Stellite 6 using the High-Velocity Oxygen Fuel (HVOF) process offers a balanced solution. **Table 2.1** summarizes common pantograph strip materials, their advantages, and limitations.

The subsequent sections review research supporting the W–Cu and Stellite 6 material combination, establishing a strong foundation for this investigation.

Table 2.1: Common Materials Used in Pantograph Contact Strip

Material Type	**Advantages**	**Disadvantages**
Pure Carbon	Good arc resistance, stable contact resistance, lightweight	High wear rate, low current-carrying capacity, brittle
Copper-Impregnated Carbon	Improved current capacity and wear resistance over pure carbon	Can be prone to selective erosion of the copper phase
Tungsten-Copper (W-Cu)	Excellent arc erosion resistance, high strength, good thermal conductivity	Moderate wear resistance, can oxidize at high temperatures
Coated Systems (Proposed)	Combines substrate conductivity with coating's wear/arc resistance	Coating adhesion and long-term integrity under cycling are key challenges

2.3 HVOF-Sprayed Stellite 6: Microstructure, Adhesion, and Elevated Temperature Performance

Stellite 6, a cobalt-chromium-tungsten carbide superalloy, is highly regarded for its exceptional wear resistance, which arises from the presence of hard, wear-resistant carbides embedded within a tough, corrosion-resistant cobalt-based matrix. The High-Velocity Oxygen Fuel (HVOF) thermal spray process is particularly effective for depositing Stellite 6 coatings. Its supersonic particle velocities produce coatings with extremely low porosity, high cohesive strength, and excellent adhesion to the substrate, while also minimizing oxidative decomposition of the feedstock powder. Recent research by [1] has evaluated the high-temperature wear performance of HVOF-sprayed Stellite 6, showing that its protective characteristics remain stable even at elevated temperatures—an essential feature for pantograph applications where frictional and arcing heating can generate localized hot spots. The integrity of any coating system largely depends on its bond with the substrate. The work of [2] highlights that laser texturing the substrate before HVOF deposition enhances this bond by creating micro-mechanical interlocking sites and increasing the effective surface area for adhesion. This process significantly improves coating durability and mitigates the risk of spallation under cyclic loading conditions, which are common in pantograph systems. Such improvements in surface engineering directly translate into longer operational life and reduced maintenance needs in high-speed rail environments

A comparative study by [3] positions HVOF-sprayed Stellite 6 as a superior coating option when compared to other deposition techniques such as Gas Tungsten Arc Welding (GTAW) cladding, demonstrating an optimal balance of wear and corrosion resistance. Findings from [4] confirm its excellent resistance to chlorine-induced corrosion, indicating potential durability against environmental contaminants like de-icing salts and industrial pollutants. Furthermore, [5] emphasizes the microstructural stability of HVOF-sprayed Stellite 6 under repeated thermal cycling, confirming that the dense lamellar structure and uniform carbide distribution remain intact even after prolonged exposure to high temperatures. This thermal resilience is crucial in ensuring consistent coating performance during pantograph operation. **Table 2.2** represents a comparative overview of various thermal spraying processes, highlighting their particle velocity, temperature, porosity, bond strength, and typical applications, thereby illustrating the advantages of the HVOF process over other coating methods. The comprehensive analysis by [6] consolidates these advantages, reporting the coating's high hardness, low oxide content, and robust microstructural integrity—properties that collectively ensure a long service life and reliable performance under the demanding conditions faced by high-speed pantograph systems.

Table 2.2 Comparison of Thermal Spray Coating Processes

Process	**Particle Velocity (m/s)**	**Temperature (°C)**	**Porosity (%)**	**Bond Strength (MPa)**	**Typical Applications**
Flame Spray	40–100	2500–3000	10–20	20–30	General coatings
HVOF	500–800	2500–3200	<2	70–80	Wear-resistant coatings
Plasma Spray	200–400	8000–15000	5–15	40–60	Ceramic coatings
Arc Spray	100–200	4000–6000	10–15	30–40	Large-area coatings

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2.4 Advancements in Deposition Techniques and Composite Modifications of Stellite 6

Although the High-Velocity Oxy-Fuel (HVOF) process is well-established for depositing Stellite 6, continuous research seeks to optimize deposition parameters and enhance coating properties through composite engineering and alternative processes. The study by [7] explored predictive modeling of cutting parameters for HVOF-sprayed Stellite 6, which is relevant for machining coated strips to meet precise geometric and surface finish requirements. Laser cladding, a viable alternative, offers metallurgical bonding and high mechanical integrity. Research by [8] on high-power laser cladding emphasized the importance of achieving defect-free coatings, setting a benchmark for quality.

To elevate performance, composite and alloy modifications have been introduced. For example, [9] developed ZrO_2-reinforced Stellite 6 coatings using laser cladding, achieving enhanced high-temperature wear resistance. Surface alloying techniques like Laser Surface Alloying (LSA) have also proven beneficial; [10] demonstrated that alloying with rhenium significantly improves wear and corrosion resistance. Furthermore, [11] showed that laser metal deposition parameters critically influence coating properties, demanding precise control. Finally, [12] introduced data-driven optimization using hybrid deep belief networks, signaling a shift toward AI-assisted material engineering to optimize coating-substrate systems for pantograph applications.

2.5 Tungsten-Copper Alloys as a Substrate: Thermal, Mechanical, and Corrosion Characteristics

The Tungsten-Copper (W-Cu 80/20) alloy substrate combines the high melting point and arc erosion resistance of tungsten with the excellent thermal and electrical conductivity of copper. This pseudo-alloy structure makes it particularly suitable for electrical contact applications, where heat dissipation and mechanical strength are critical. According to [17], W-Cu composites efficiently dissipate heat, preventing thermal softening and maintaining integrity under severe operating conditions. The high-temperature tribological studies by [16] confirmed that while W-Cu alloys exhibit moderate wear resistance, a protective coating is essential to prevent adhesive and abrasive wear.

Mechanical behavior under extreme conditions further validates their application potential. The study by [18] on W-Cu alloys under high-strain-rate loading highlights their dynamic strength, while [14] demonstrated that grain refinement enhances both mechanical properties and surface quality for improved coating adhesion. Corrosion studies by [15] indicated vulnerability in saline environments, reinforcing the need for protective coatings. Additionally, [13] explored helium ion irradiation effects, showing microstructural stability under radiation exposure—an indication of the alloy's robustness for long-term, high-stress operations.

2.6 Operational Challenges and Degradation Mechanisms in Pantograph-Catenary Systems

High-speed pantograph systems face a complex interplay of degradation mechanisms, where mechanical wear, electrical arcing, and environmental factors converge. Electrical arcing during intermittent contact is a primary cause of material loss and surface erosion. The work of [19] investigated arc erosion under varying humidity conditions, establishing a direct link between atmosphere and damage severity. Simulation studies by [21] modelled plasma temperatures in DC pantograph arcs, predicting contact line ablation and quantifying thermal stresses experienced by coating materials.

The influence of current on tribological performance, or tribo-electrical behaviour, is equally critical. Studies by [20] and [22] demonstrated that electrical current accelerates wear through localized melting and oxidation. System reliability studies, such as [23], applied decision tree models to predict sliding strip failures, emphasizing a data-driven maintenance approach. Complementary work by [24] examined material defects as damage initiation sites, while [25] reviewed real-time monitoring technologies, underlining the growing integration of predictive analytics in ensuring pantograph system safety and performance.

2.7 Benchmarking Against Alternative Coating: WC-Based and Other Cermet Systems

Comparative benchmarking is essential to justify Stellite 6 selection over alternatives like tungsten carbide (WC)-based coatings. WC-Co-Cr coatings are widely recognized for their hardness and wear resistance. According to [26], the binder composition critically affects microstructure and performance in HVOF-sprayed WC-Co-Cr coatings. The study by [27] compared cold-sprayed and HVOF-sprayed WC-10Co4Cr coatings, revealing process-dependent differences in microstructure and mechanical behavior. Such comparisons underscore that the deposition process influences the final coating performance as much as the material itself.

Further insights from [28] highlight issues like decarburization and brittle phase formation that can compromise WC-cermet coatings. The study by [29] on WC-12Co coatings revealed performance sensitivity to load and counter-body material—factors significant for pantograph applications. While WC coatings excel in hardness, Stellite 6 offers better toughness, corrosion resistance, and galling resistance under metal-to-metal sliding. Emerging materials like ZrCN coatings studied by [30] represent future possibilities, but currently, Stellite 6 remains unmatched for applications demanding both toughness and tribological resilience.

7. Synthesis and Identification of the Research Gap

The literature collectively establishes HVOF-sprayed Stellite 6 and W-Cu alloys as high-performance materials for severe service conditions. Studies [1], [3], [4], [5], and [6] confirm Stellite 6's excellent combination of wear resistance, corrosion protection, and microstructural stability. Meanwhile, [16], [17], and [18] underscore W-Cu's superior conductivity, arc resistance, and mechanical durability, making it an ideal substrate. The interplay of degradation mechanisms—including arc erosion [19], [21], electrical wear [20], [22], and environmental effects [15]—is well understood, as is the comparative performance of alternative coatings [26]– [29].

However, a significant research gap persists in the current body of literature. While numerous studies have independently focused on improving the tribological or electrical characteristics of contact materials, there is a noticeable absence of comprehensive investigations that **synergistically integrate the wear-resistant Stellite 6 coating with a thermally conductive W–Cu substrate**, particularly for **pantograph–catenary applications** in high-speed railway systems. Most existing works emphasize either coating optimization or substrate selection in isolation, without addressing their combined influence on contact performance under dynamic service conditions such as high sliding speeds, fluctuating contact pressures, and varying environmental factors.

The combined tribological, electrical, and thermal performance of this hybrid system under realistic operating conditions remains largely unexplored. Understanding the interfacial behavior between Stellite 6 and W–Cu, especially in terms of adhesion strength, microstructural compatibility, and heat dissipation during continuous current transfer, is essential for ensuring long-term operational stability. Therefore, this project seeks to bridge this critical research gap by experimentally evaluating HVOF-sprayed Stellite 6 coatings on W–Cu substrates. Through detailed characterization and performance testing, the study aims to establish a functionally graded surface-engineered system that enhances wear resistance, minimizes contact losses, and ensures efficient heat transfer. This integrated approach offers a novel and practical solution to improve the durability, efficiency, and reliability of next-generation high-speed railway pantograph systems, contributing significantly to sustainable and maintenance-free rail transport technologies.

CHAPTER 3: PROBLEM IDENTIFICATION

The drive for higher operational speeds, improved efficiency, and greater reliability in modern railway systems has exposed significant weaknesses in existing pantograph contact strip technologies. Traditional monolithic materials such as copper, carbon, and steel are increasingly unable to withstand the demanding conditions of high-speed rail operations. Their limited-service life and inconsistent performance under high current density, fluctuating contact forces, and elevated temperatures have become critical challenges. These issues arise not from isolated failures but from an overall mismatch between material capabilities and modern operational requirements. Consequently, performance losses, unstable current collection, and premature wear lead to reduced reliability and higher energy losses, highlighting the need for advanced material innovation.

A key limitation lies in the intrinsic property trade-off within monolithic materials. Metals like copper and silver provide excellent electrical conductivity, ensuring efficient current transfer, but are mechanically soft and wear quickly under continuous contact and arcing. Conversely, materials such as carbon, ceramics, and carbides exhibit good wear resistance and thermal stability but suffer from poor electrical conductivity. This unavoidable conflict forces designers to compromise between electrical efficiency and durability.

In addition, pantograph strips are subjected to multi-mode degradation, including mechanical abrasion, adhesion, oxidation, and arc erosion. For instance, carbon strips resist wear but deteriorate under high arc energy, while copper strips conduct well but deform or melt at elevated temperatures. The severe consequence of these combined mechanisms is vividly illustrated in **Figure 3.1**, which shows a heavily worn pantograph contact strip. The visible grooving, pitting, and material loss are direct results of the relentless mechanical sliding and intermittent electric arcing. This level of degradation not only necessitates frequent replacement but also increases the risk of catastrophic failure and damage to the overhead catenary wire. These coupled degradation mechanisms are summarized in **Table 3.1**, which outlines specific material problems and their resulting consequences on performance and service life.

As train speeds now exceed 300 km/h and traction power demands rise, the severity of these conditions intensifies. Increased sliding velocity and higher current densities accelerate heating and arc damage, leading to the irregular wear patterns evident in **Figure 3.1**. This results in unstable contact resistance and inefficient power transmission. These limitations translate into high life-cycle costs increased downtime, and potential catenary wire damage — all of which reduce system efficiency and reliability.

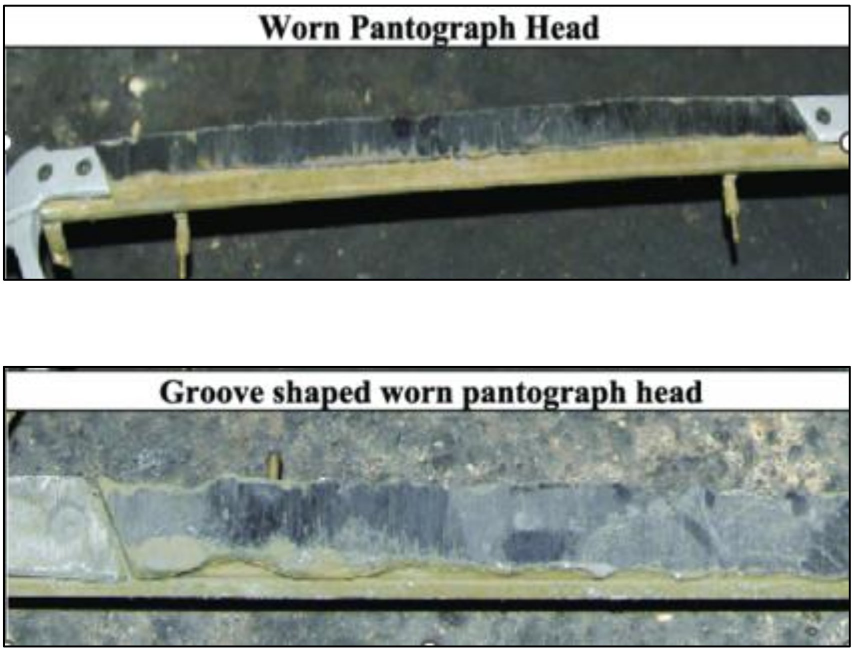


Figure 3.1: Wear in Pantograph

Therefore, **Table 3.1** emphasizes the urgent need for advanced material solutions such as, tungsten–copper substrates, or the graded hybrid structure proposed in this research that can balance conductivity, wear resistance, and thermal performance. The development of such next-generation materials will be essential to achieve longer service life, improved energy efficiency,

Table 3.1: Summary of Identified Problems and Their Consequences

Identified Problem	Consequence
Property Trade-off: No single material offers both high conductivity and high wear resistance.	**Performance Limitation**: Either good electrical efficiency or durability, not both.
Multi-Mode Degradation: Contact strips face wear, arc erosion, and thermal fatigue simultaneously.	**Reduced Life**: Leads to rapid material loss, cracks, and surface damage.
High Operational Demand: Increased speed and load cause greater thermal and mechanical stress.	**Early Failure**: Materials reach performance limits faster under modern conditions.
Frequent Maintenance Needs: Regular replacement of worn strips.	**Higher Cost**: Increases downtime and maintenance expenses.

CHAPTER 4: OBJECTIVES

The primary aim of this research is to conceptualize, develop, and initiate the experimental validation of a novel, surface-engineered pantograph contact strip designed to overcome the limitations of traditional monolithic materials. The goal is to create a functionally graded composite system that synergistically combines the favorable bulk properties of a tungsten-copper substrate with the exceptional surface properties of a Stellite 6 coating, thereby achieving a significant enhancement in service life, operational reliability, and performance for high-speed rail applications. This overarching aim is broken down into the following specific, measurable, and sequentially structured objectives to provide a clear roadmap for the investigation.

1. **System Design and Specification:** To design the composite system by selecting a tungsten-copper (W-Cu) alloy with a composition of 80% tungsten and 20% copper by weight as the substrate, and Stellite 6 as the coating material, with deposition to be carried out using the High-Velocity Oxygen Fuel (HVOF) process.

2. **Specimen Fabrication and Preparation:** To procure a certified W-Cu 80/20 alloy and fabricate a set of test specimens with specific geometries (30 mm, 5 mm, and 2 mm in length) using Wire Electrical Discharge Machining (Wire EDM) to ensure dimensional accuracy and surface integrity.

3. **Coating Application:** To prepare the substrate surfaces and deposit a dense, adherent Stellite 6 coating using the optimized HVOF thermal spray process, with careful control and documentation of all process parameters.

4. **System Evaluation and Characterization:** To conduct a comprehensive evaluation of the composite system, including:
    - Microstructural analysis (coating morphology, thickness, porosity, and interface integrity) using Optical Microscopy, SEM, and EDX.
    - Mechanical and tribological testing (microhardness, adhesion strength, wear rate, and coefficient of friction).
    - Functional performance testing (electrical conductivity and arc erosion resistance).

5. **Comparative Analysis:** To quantitatively compare the performance of the HVOF-coated Stellite 6 on W-Cu against uncoated W-Cu and other benchmark materials to demonstrate the performance enhancement achieved.

In addition to the outlined objectives, this research further extends into process optimization, computational modeling, and life-cycle evaluation to ensure a comprehensive understanding of the system's performance. The optimization of process parameters focuses on examining the effects of HVOF spray conditions—such as oxygen-to-fuel ratio, spray distance, and particle velocity—on coating quality, adhesion, and porosity. Establishing an optimal parameter window will be essential to achieve consistent coating integrity and reproducibility for large-scale applications.

CHAPTER 5: PROBLEM SOLUTION

The identified problems of property trade-offs and multi-mode degradation necessitate a solution that moves beyond the paradigm of monolithic materials. The proposed solution, central to this research, is a surface engineering approach that creates a functionally graded composite system.

This strategy deliberately decouples the functional requirements of the bulk material from those of the surface, allowing for the optimization of each region independently to meet the conflicting demands of the application. The core of the solution lies in the strategic combination of a tungsten-copper (W-Cu 80/20) substrate with a Stellite 6 coating applied via the High-Velocity Oxygen Fuel (HVOF) thermal spray process.

The W-Cu substrate is chosen to fulfil the bulk material requirements. Its composite microstructure, featuring a continuous tungsten skeleton infiltrated with copper, provides an optimal compromise. The copper phase ensures the high electrical and thermal conductivity necessary for efficient current collection and heat dissipation away from the contact zone, addressing the need for electrical efficiency. Simultaneously, the tungsten phase provides the mechanical strength, rigidity, and inherent resistance to arc erosion and deformation that pure copper lacks. This makes the substrate far more robust than traditional copper-based strips, forming a durable and conductive foundation.

The surface requirements—extreme resistance to mechanical wear, arc erosion, and thermal softening—are addressed by the Stellite 6 coating. This cobalt-based superalloy, with its hard chromium carbide precipitates embedded in a tough, oxidation-resistant cobalt-chromium matrix, is specifically engineered to withstand severe tribological conditions. When applied via the HVOF process, it forms a dense, well-bonded shield on the substrate surface. The HVOF process is critical as its high particle velocity results in a coating with minimal porosity and high adhesion, ensuring the protective layer remains intact under operational stresses.

The synergy of this material pair is the key innovation. The Stellite 6 coating protects the underlying W-Cu substrate from the direct onslaught of wear and arcing, drastically reducing material loss. In turn, the W-Cu substrate provides the structural and electrical backbone, efficiently conducting current and heat, and preventing the kind of catastrophic failure that might occur in a brittle, standalone ceramic or carbide coating. This solution directly tackles the identified problems: it overcomes the property trade-off by using one material for conduction (substrate) and another for wear resistance (coating); it resists multi-mode degradation as both materials are chosen for their complementary strengths against different damage mechanisms; and by significantly extending service life, it promises a substantial reduction in the life-cycle cost associated with pantograph contact strip maintenance and replacement.

CHAPTER 6: BLOCK DIAGRAM

A systematic and sequential methodology is essential for the successful execution of this research project. The entire investigation, from problem identification to the final conclusion, can be visually summarized and logically structured using a block diagram, as presented in **Figure 6.1**. This flowchart outlines the step-by-step process, ensuring a coherent and organized approach.

The process begins with the initial phase of Problem Identification and Literature Review, which establishes the need for the research and the current state of knowledge. This foundational work directly informs the formulation of the project's Aim and Objectives, providing a clear and focused direction for all subsequent activities.

The first major experimental stage is Material Selection and Specimen Preparation. This block involves the procurement of the certified tungsten-copper (W-Cu) substrate material, followed by the detailed design and precision machining of the test specimens using Wire EDM. The successful completion of this stage yields a set of characterized and ready-to-coat specimens.

The next critical stage is Coating Application and Process Optimization. This involves the preparation of the specimen surfaces (e.g., grit blasting) and the deposition of the Stellite 6 coating using the HVOF thermal spray process. Parameters may be varied and optimized to achieve the desired coating quality.

The core of the investigative work is encapsulated in the Testing and Characterization block. This is a multi-faceted stage where the coated composites undergo a battery of tests, including microstructural analysis (OM, SEM, EDX), mechanical testing (hardness, adhesion), tribological testing (wear rate), and functional testing (electrical conductivity, arc erosion). Parallel to this, uncoated reference specimens are tested to establish a baseline for comparison.

The data generated from all characterization and testing phases are then collected for Comparison. This involves processing the raw data, calculating performance metrics, and systematically comparing the results of the coated specimens against the uncoated baseline and other relevant benchmarks from the literature.

The interpreted results lead to the conclusion and documentation phase, where the findings are summarized, the initial hypotheses are validated or refuted, the project objectives are reviewed, and the entire research is compiled into the final report and potential scientific publications. This block diagram serves as a vital roadmap, ensuring methodological rigor and clarity throughout the project's lifecycle.

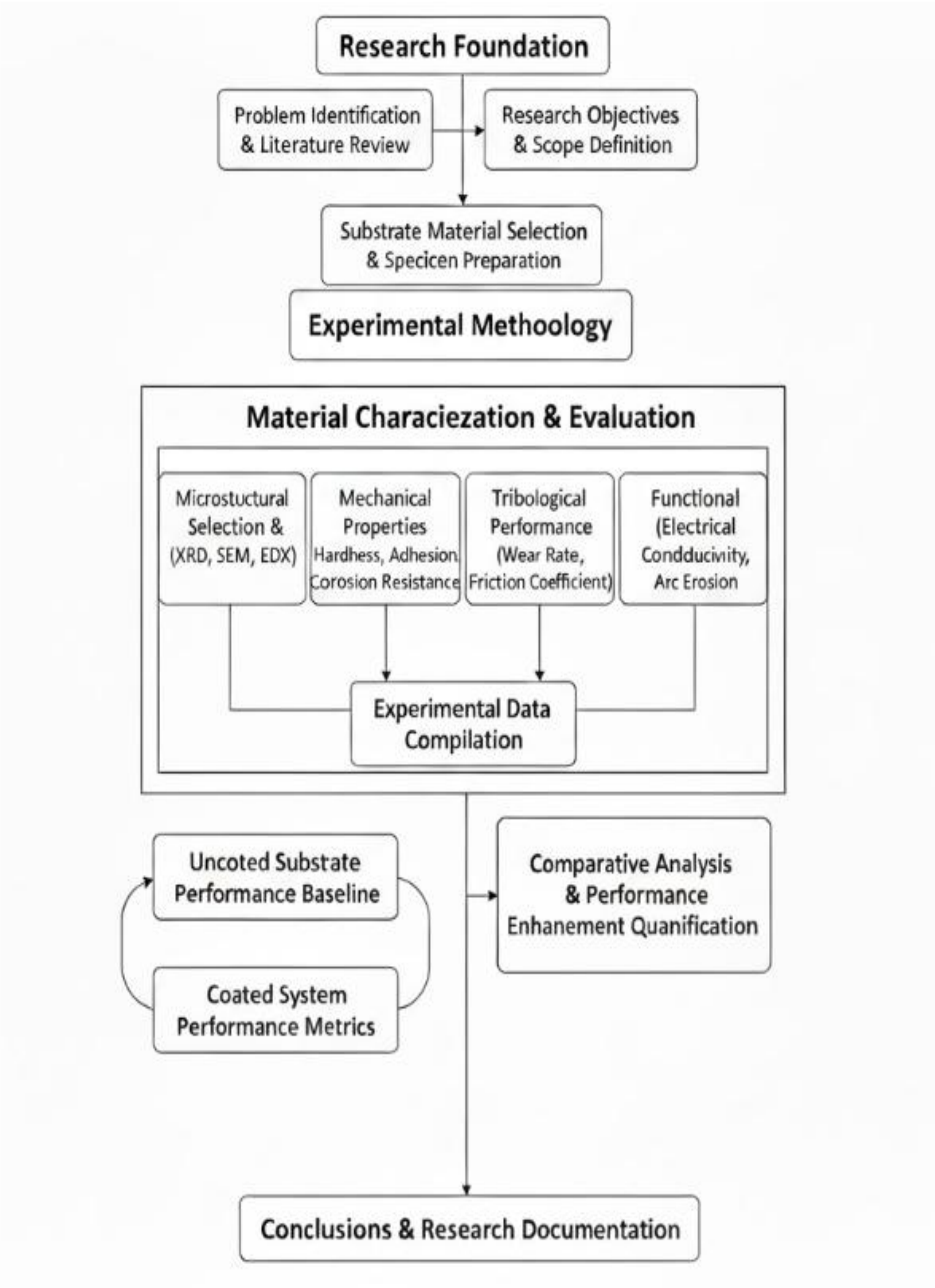


Figure 6.1: Research Methodology Block Diagram

CHAPTER 7: METHODOLOGY

7.1 Substrate Material: Tungsten-Copper 80/20 Alloy

The foundation of the proposed composite system is a tungsten–copper (W–Cu) pseudo-alloy with a nominal composition of 80% tungsten and 20% copper by weight. This material is not a true alloy due to the mutual insolubility of tungsten and copper in both the solid and liquid states; instead, it is a composite produced through powder metallurgy techniques.

The manufacturing process involves pressing and sintering high-purity tungsten powder to form a porous skeletal structure, which is subsequently infiltrated with molten copper under a controlled atmosphere. The resulting material is fully dense and exhibits a distinct microstructure of tungsten particles embedded in a continuous copper matrix, as illustrated in **Figure 7.1**.

Figure 7.1: Tungsten-Copper Alloy Microstructure

The tungsten phase, characterized by its exceptionally high melting point (3422°C) and hardness, provides mechanical strength, stiffness, and wear resistance. The copper phase offers excellent electrical and thermal conductivity by forming continuous conductive paths within the structure. The 80W–20Cu composition represents an optimized balance — higher copper content improves conductivity but reduces strength, whereas higher tungsten content enhances hardness but decreases conductivity.

This grade typically provides 20–25% IACS electrical conductivity, high mechanical integrity, and superior arc erosion resistance. Its established performance in demanding applications such as EDM electrodes, resistance welding tips, and high-power electronic heat sinks further justifies its selection as the substrate material for this investigation. The chemical composition of the selected W–Cu 80/20 alloy is summarized in **Table 7.1**, and its physical and mechanical properties are presented in **Table 7.2**.

Table 7.1: Chemical Composition of W-Cu 80/20 Alloy

Element	**Composition (wt%)**	**Role in the Alloy**
Tungsten (W)	80 ± 2	Provides hardness, wear resistance, and high-temperature strength
Copper (Cu)	20 ± 2	Provides electrical and thermal conductivity, ductility
Impurities	< 0.5	Trace elements from the manufacturing process

Table 7.2: Physical and Mechanical Properties of W-Cu 80/20 Alloy

Property	Value (Min–Max)	Typical Value	Unit
Density	15.3 – 15.8	15.6	g/cm³
Electrical Conductivity	28 – 32	30	% IACS
Thermal Conductivity	130 – 150	140	W/m·K
Hardness	220 – 260	240	HV
Tensile Strength	610 – 760	680	MPa
Elastic Modulus	230 – 250	240	GPa

7.2 Coating Material: Stellite 6

The coating material used in this study is Stellite 6, a cobalt-based superalloy well known for its excellent wear, corrosion, and oxidation resistance over a wide temperature range. Unlike conventional alloys, its properties arise from its chemical composition and stable microstructure rather than heat treatment.

Stellite 6 mainly consists of cobalt (base element), chromium (28–32%), tungsten (4–6%), and carbon (1.0–1.4%), with small amounts of nickel, iron, and silicon are represented in **Table 7.3**. Cobalt provides toughness and high-temperature strength; chromium enhances corrosion resistance and forms protective carbides; tungsten improves strength at elevated temperatures; and carbon enables the formation of hard carbide phases. Minor elements improve fluidity and stability during processing.

The microstructure of Stellite 6 is represented in **Figure 7.2** consists of a cobalt-rich solid solution matrix reinforced by hard chromium carbides (Cr_7C_3, $Cr_{23}C_6$), which occupy about 15–30% of the structure and possess hardness values exceeding 1500 HV.

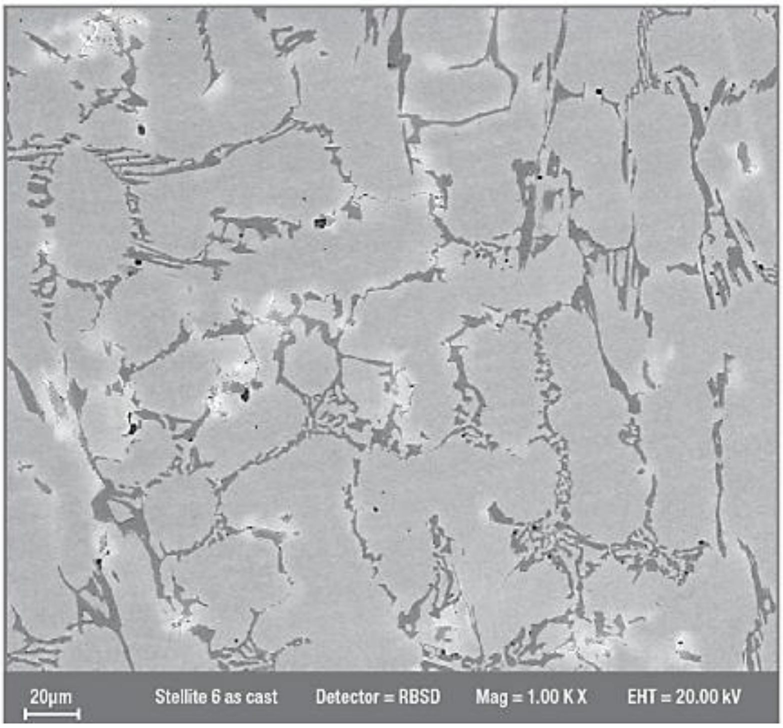


Figure 7.2: Stellite 6 Microstructure with Carbide Phases

For HVOF (High Velocity Oxy-Fuel) applications, Stellite 6 is used as a gas-atomized powder with good flowability and uniform melting characteristics. HVOF-sprayed coatings typically exhibit hardness between 350–450 HV, strong substrate adhesion, and low porosity, ensuring durable surface protection for the tungsten–copper substrate in pantograph contact applications.

Additionally, Stellite 6 maintains its mechanical and chemical stability even under high friction and thermal cycling conditions, making it suitable for components exposed to intermittent arcing and sliding wear.
Furthermore, its proven track record in aerospace, turbine, and tribological systems justifies its selection for this research as a reliable and high-performance coating material.

Table 7.3: Chemical Composition of Stellite 6

Element	Composition (wt%)	Role in the Alloy
Cobalt (Co)	Balance	Matrix material providing toughness and high-temperature strength
Chromium (Cr)	28-32	Provides corrosion resistance and forms hard carbides
Tungsten (W)	4-6	Solid solution strengthener and carbide former
Carbon (C)	1.0-1.4	Essential for carbide formation
Nickel (Ni)	≤ 3	Austenite stabilizer and solid solution strengthener
Iron (Fe)	≤ 3	Impurity from raw materials

7.3 Coating Process: HVOF Technology

The High-Velocity Oxygen Fuel (HVOF) thermal spray process was selected as the coating application method for this project due to its proven capability to produce coatings of exceptional density, strength, and quality, which are critical for the demanding application of a pantograph contact strip.

As illustrated in **Figure 7.3**, the HVOF process operates on the principle of continuous internal combustion of a fuel gas with oxygen within a specialized spray gun. Common fuel gases include propylene, hydrogen, or kerosene. This combustion occurs in a chamber at high pressure—typically several times atmospheric pressure—generating a high-temperature (2500–3200°C), high-pressure gas stream. The gases are expelled through a converging-diverging (de Laval) nozzle, accelerating to supersonic velocities.

The coating material is fine Stellite 6 powder, is injected axially into this high-velocity gas stream, where it is rapidly heated to a molten or semi-molten state and propelled toward the prepared substrate at speeds ranging from 500 to 800 m/s. Upon impact, the particles flatten plastically into thin splats and solidify rapidly, building the coating layer by layer through successive deposition.

The key advantages of the HVOF process stem from its high particle velocity and short residence time in the flame. These result in coatings with extremely low porosity (<2%), superior adhesion, minimal oxidation, and excellent microstructural integrity. Additionally, the induced compressive residual stresses improve the coating's fatigue and spallation resistance.

Overall, the HVOF process provides an ideal platform for depositing the Stellite 6 coating on the W–Cu substrate, ensuring high coating density, strong interfacial bonding, and enhanced wear and oxidation resistance—attributes essential for reliable pantograph operation.

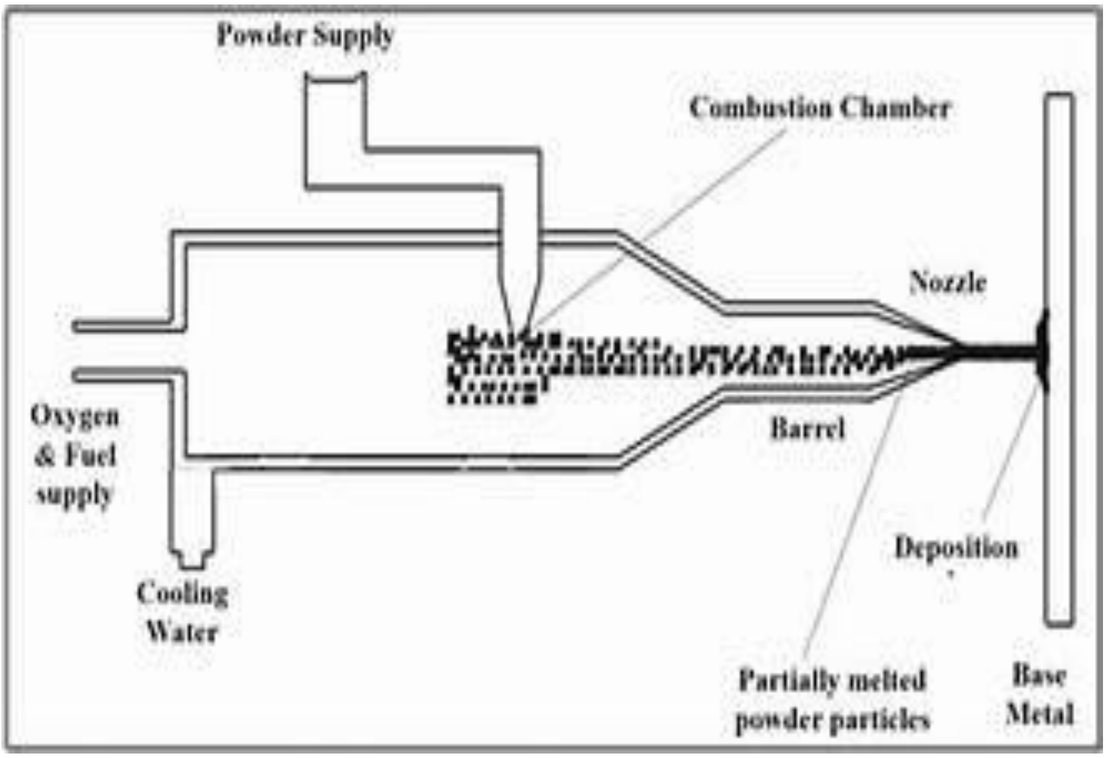


Figure 7.3: Schematic of the HVOF Spray Process

7.4 Specimen Design and Rationale

The design of the test specimens was a critical step that required careful consideration of multiple, often competing, requirements to ensure they would be fit for their intended purposes throughout the experimental program. The primary constraints included the need for sufficient surface area for coating application and subsequent testing, compatibility with available testing equipment and characterization tools, conservation of the expensive substrate material, and ease of handling. After a thorough review of these factors, a cylindrical geometry was selected, maintaining the as-received diameter of 10 mm from the procured rod. This diameter provides an adequate surface area for depositing a uniform HVOF coating and is compatible with standard specimen holders for tribological testers and microscopy stages. To address the different requirements of various tests, three distinct specimen lengths were designed, as detailed in **Table 7.4.**

- **Standard Test Specimen (30 mm):** These longer specimens provide ample material for secure fixturing in mechanical test rigs, such as those for wear testing and bond strength evaluation. Their length allows a portion to be gripped firmly in a collet or fixture while leaving a sufficient exposed surface for creating wear tracks or attaching pull-off adhesives without interference from the edges.
- **Characterization Specimen (5 mm):** These short, disc-shaped specimens are ideally suited for metallographic preparation. Their minimal thickness allows for efficient cross-sectional mounting, grinding, and polishing to reveal the coating-substrate interface and the coating's internal microstructure for examination under optical and electron microscopes.
- **Corrosion Test Specimen (2 mm):** These were designed to provide a flat, coated surface for electrochemical analysis while minimizing material usage.

Using shorter specimens for destructive analysis conserves material. A total of sixteen specimens were planned: four of the 30 mm length, eight of the 5 mm length, and four of the 2 mm length. This quantity provides for multiple coating trials, replicates for statistical reliability in testing, and includes uncoated reference specimens for baseline comparison. This multi-specimen approach ensures that all planned characterization and testing activities can be conducted with appropriately designed samples, thereby maximizing the scientific value derived from the procured material while maintaining efficiency and practicality.

Table 7.4: Specimen Size Specifications and Purpose

Specimen Type	Length (mm)	Diameter (mm)	Quantity	Primary Use
Standard Test Specimen	30 ± 0.2	10 ± 0.02	4	Mechanical & Tribological Testing (e.g., Pin-on-Disc wear tests, bond strength adhesion tests)
Characterization Specimen	5 ± 0.2	10 ± 0.02	8	Microstructural Analysis
Corrosion Test Specimen	2 ± 0.2	10 ± 0.02	4	Electrochemical and Corrosion Analysis

7.5 Machining Process: Wire EDM

The selected method for dividing the procured tungsten-copper rod into the designed specimens was Wire Electrical Discharge Machining (Wire EDM). This non-conventional machining process was chosen over conventional methods like sawing or abrasive cutting due to its unique advantages when working with hard, brittle, and electrically conductive materials like tungsten-copper.

Conventional machining techniques impose significant mechanical forces on the workpiece, which can lead to micro-cracking, edge chipping, or delamination in composite or brittle materials. They also cause rapid tool wear when cutting hard substances. Wire EDM circumvents these issues entirely by utilizing a thermoelectric process to remove material. The process involves a thin, continuously moving wire electrode (typically brass or coated brass with a diameter of 0.1-0.3 mm) and the workpiece, both submerged in a dielectric fluid, usually deionized water. A controlled electrical potential is applied between the wire and the workpiece, creating a series of rapid, repetitive spark discharges across a small gap. Each spark generates an intense, localized plasma channel with temperatures around 8,000-12,000°C, instantly melting and vaporizing a microscopic amount of material from both the wire and the workpiece. The flowing dielectric fluid serves to flush away the eroded particles, cool the workpiece, and deionize the spark gap to prevent arcing.

The wire is guided along a programmed CNC path, allowing for precise and complex cuts. For this project, the path was simply a straight line through the 10 mm diameter of the rod to produce perpendicular cuts. The key benefits of using Wire EDM for this application are manifold. It imposes no direct mechanical cutting forces, thereby eliminating the risk of mechanical damage, stress, or cracking in the brittle tungsten-copper material. It provides exceptional dimensional accuracy and repeatability, with tolerances easily held within ±0.025 mm. It produces a good surface finish directly from the machining process, characterized by a fine, recast layer texture. The cut is perfectly perpendicular to the rod axis with minimal deviation. Furthermore, the heat-affected zone is extremely shallow and localized, preventing any significant thermal alteration of the bulk material's properties. The specific parameters used are detailed in **Table 7.5**. This combination of precision, gentle material removal, and excellent surface integrity made Wire EDM the unequivocal choice for preparing the high-quality specimens required for this sensitive materials investigation.

Table 7.5: Wire EDM Process Parameters

Parameter	Setting	Unit
Wire Material	Brass (Cu-Zn)	-
Wire Diameter	0.25	mm
Pulse On Time	8	µs
Pulse Off Time	40	µs
Peak Current	180	A
Wire Tension	12	N
Dielectric Fluid	Deionized water	-
Conductivity	<10	µS/cm
Flushing Pressure	1.2	MPa

7.6 Experimental Execution: Material Procurement and Verification

The successful initiation of the experimental work hinged on the procurement of a high-quality, certified tungsten-copper (W–Cu) 80/20 alloy substrate material. After evaluating multiple specialized suppliers of refractory metals and advanced composites, a purchase order was placed with a reputed vendor for a certified W–Cu 80/20 alloy rod measuring 200 mm in length and 10 mm in diameter.

As shown in **Figure 7.4**, the as-received tungsten-copper rod exhibited a uniform, silver-gray metallic luster with the characteristic high density of the material. A preliminary visual inspection under proper lighting confirmed the absence of macroscopic defects such as cracks, scratches, or surface pits. The cylindrical surface displayed fine grinding marks, while the ends retained a slightly rough finish from the manufacturer's cutting process.

Dimensional verification was performed using a calibrated digital caliper (accuracy ±0.01 mm). Multiple measurements taken along the length and circumference confirmed excellent uniformity, with diameters ranging between 9.98 mm and 10.02 mm, and a total length of 201 mm, which was within acceptable tolerance.

The material was supplied with a Certificate of Analysis, verifying its composition and key physical properties, summarized in Table 7.6. Optical emission spectroscopy confirmed that tungsten and copper contents met the nominal 80/20 ratio, with impurities below 0.1%. The measured density of 15.7 g/cm^3 (via Archimedes method) indicated a fully dense structure with negligible porosity, and an electrical conductivity of 22.5% IACS, consistent with the expected range for this grade.

This certification, combined with the successful visual and dimensional assessments, ensured that the substrate met all technical requirements. The material was therefore accepted and formally logged into the project inventory, marking the successful completion of the procurement and verification phase.

Figure 7.4: As-Received Tungsten-Copper Rod

Table 7.6: Material Certification Data

Property	Specification	Measured Value	Test Method
Chemical Composition			Optical Emission Spectroscopy
- Tungsten	80 wt.% (nominal)	80.2 wt.%	
- Copper	20 wt.% (nominal)	19.8 wt.%	
- Total Impurities	< 0.5 wt.%	< 0.1 wt.%	
Density	15.6 - 15.8 g/cm^3	15.7 g/cm^3	Archimedes Method
Electrical Conductivity	20 - 25 % IACS	22.5 % IACS	Eddy Current
Manufacturing Method	-	Powder metallurgy with liquid copper infiltration	

7.7 Experimental Execution: Wire EDM Specimen Preparation

With the certified tungsten–copper rod in hand, the next critical step was its precision segmentation into test specimens using the Wire EDM process. As illustrated in **Figure 7.5**, Wire Electrical Discharge Machining (EDM) is a non-contact thermal erosion process that utilizes a continuously moving, electrically charged wire electrode to cut conductive materials with exceptional accurac

Figure 7.5: Wire Cut EDM Process

The machining was carried out on a modern CNC-controlled Wire EDM machine. Prior to cutting, the rod was thoroughly cleaned with acetone to remove surface contaminants that could interfere with spark discharge and was then securely mounted in the machine's vice to ensure firm yet distortion-free clamping.

The cutting program was configured to produce four specimens of 30 mm length, eight specimens of 5 mm length, and four specimens of 2 mm length. A preliminary trial cut on a scrap piece was used to fine-tune key process parameters—pulse on/off time, discharge current, and wire feed rate—to achieve optimal machining conditions for the tungsten–copper composite

.

During cutting, a brass wire electrode traversed through the 10 mm-diameter rod while submerged in deionized water, which acted as the dielectric medium. Controlled spark discharges progressively eroded the material along the programmed path. Each 30 mm specimen required approximately 40 minutes, while the 5 mm and 2 mm pieces took 28 minutes and 20 minutes, respectively.

Throughout the operation, machining stability was continuously monitored, and dielectric conductivity was maintained by periodic water replacement. The process ran smoothly, with only one minor interruption—a temporary reduction in dielectric flow caused by a partially clogged filter—which was quickly corrected without affecting part quality.

Upon completion, all sixteen specimens were cleanly separated with precise dimensions, sharp edges, and no visible surface damage, confirming the suitability of the Wire EDM process for fabricating high-precision W–Cu test samples.

Additionally, dimensional accuracy of the machined specimens was verified using a digital micrometre and a coordinate measuring machine (CMM), confirming deviations within ±0.01 mm tolerance. Surface roughness was also measured using a contact profilometer, revealing an average Ra value below 1.2 µm, demonstrating the capability of Wire EDM to achieve high-quality finishes on difficult-to-machine composite materials. A concise summary of the operation parameters is presented in **Table 7.7**.

Table 7.7: Wire EDM Cutting Summary

Cutting Sequence	Specimen Type	Target Length (mm)	Quantity	Average Cutting Time (min)	Observations
1-4	Standard Test	30	4	40	Stable cutting, good surface finish
5-12	Characterization	5	8	28	Faster cutting due to reduced thickness
13-16	Corrosion Test	2	4	20	Stable cutting, good surface finish

7.8 Experimental Execution: Specimen Quality Assessment

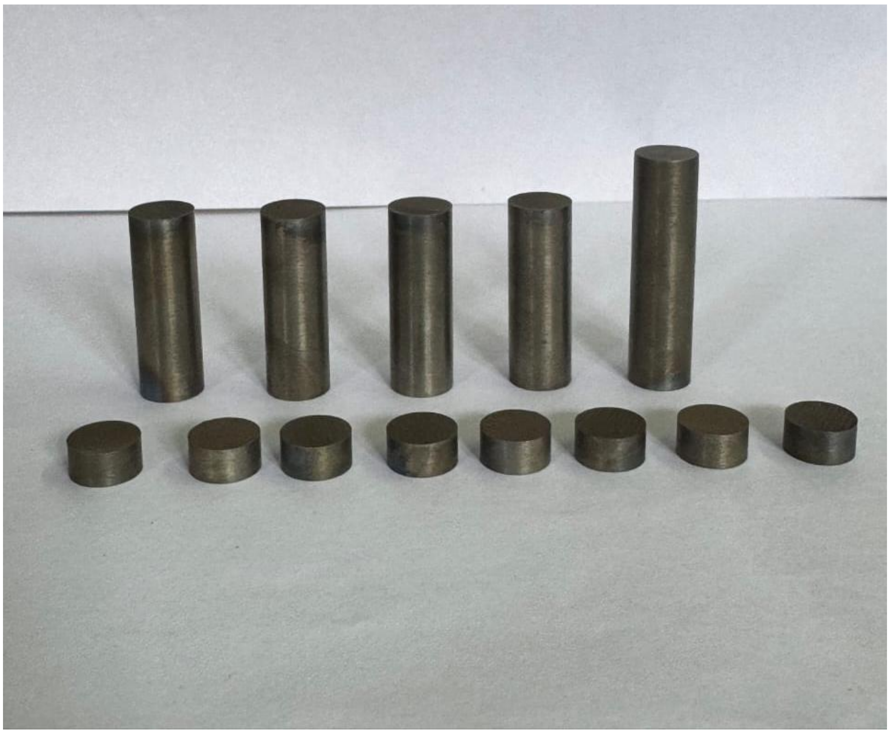
Following the Wire EDM process, all sixteen specimens underwent a comprehensive quality assessment to verify that they met the stringent requirements for subsequent coating and testing. The assessment protocol included dimensional verification, mass measurement, visual and tactile surface examination, and a check for structural integrity.

Each specimen was measured using digital calipers to confirm its length and diameter. The results, detailed in Table 7.8, demonstrated excellent dimensional control. The four 30 mm specimens showed lengths within ±0.15 mm of the target, while the eight 5 mm and four 2 mm specimens exhibited slightly greater variation, as expected for thinner cuts, but all were well within the acceptable ±0.2 mm tolerance. The diameter remained consistently at 10.00 ± 0.02 mm across all specimens, indicating no measurable ovality or tapering.

The mass of each specimen was measured using a precision analytical balance. The masses aligned perfectly with the theoretical mass calculated from the dimensions and the certified density of 15.7 g/cm³, providing further confirmation of the material's full density and the accuracy of the cutting process.

Visual inspection of the cut surfaces under ambient light and with a 10x magnifying lens revealed the characteristic texture of an EDM surface: a uniform, matte-grey appearance with fine, overlapping craters from the spark discharges. No evidence of cracking, chipping, or gross defects was observed on any specimen. The edges, where the EDM cut intersected the original cylindrical surface, were sharp and clean without any rounding or micro-fractures. Tactile inspection by gently running a fingernail across the surfaces confirmed a consistent texture without rough patches. The structural integrity was further verified by a simple tap test; each specimen produced a clear, sharp ringing sound when lightly struck, indicative of a dense, crack-free condition.

This rigorous quality assessment confirmed that all sixteen specimens were manufactured to a high standard, with dimensions and surface conditions fully suitable for the next stages of the experimental plan, which involve surface preparation and the application of the HVOF-sprayed Stellite 6 coating. The final inventory is catalogued in **Table 7.8, Table 7.9** and pictured in **Figure 7.6**.



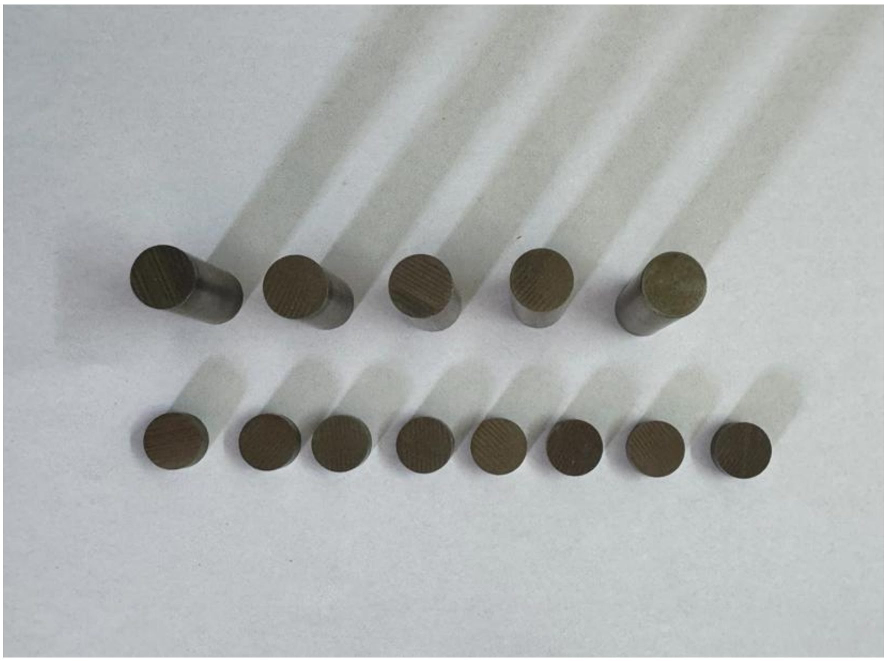


Figure 7.6: Prepared Set of Test Specimen

Table 7.8: Dimensional Measurements of Cut Specimens

Specimen ID	Length (mm)	Diameter Range (mm)	Mass (g)	Length Deviation from Target (mm)
30 mm Specimens				
W-Cu-01	30.15	9.98 - 10.01	37.14	+0.15
W-Cu-02	30.08	9.99 - 10.02	37.02	+0.08
W-Cu-03	29.92	9.98 - 10.00	36.89	-0.08
W-Cu-04	30.11	9.99 - 10.01	37.08	+0.11
5mm Specimens				
W-Cu-05	5.12	10.00 - 10.01	6.22	+0.12
W-Cu-06	4.95	9.98 - 9.99	6.17	-0.05
W-Cu-07	5.08	9.99 - 10.00	6.19	+0.08
W-Cu-08	4.98	10.01 - 10.02	6.18	-0.02
W-Cu-09	5.05	9.97 - 9.98	6.14	+0.05
W-Cu-10	4.88	9.99 - 10.00	6.16	-0.12
W-Cu-11	5.03	10.00 - 10.01	6.20	+0.03
W-Cu-12	4.92	9.98 - 9.99	6.15	-0.08
2mm Specimens				

W-Cu-13	2.05	9.99 - 10.00	2.48	+0.05
W-Cu-14	1.98	10.00 - 10.01	2.45	-0.02
W-Cu-15	2.10	9.98 - 9.99	2.46	+0.10
W-Cu-16	1.95	9.99 - 10.00	2.44	-0.05

Table 7.9: Complete Specimen Inventory

Specimen ID	Length (mm)	Diameter (mm)	Mass (g)	Designated Use	Status
Standard Test Specimens (30 mm)					
W-Cu-01	30.15	10.00	37.14	For coating and mechanical testing	Cleaned and stored
W-Cu-02	30.08	10.01	37.02	For coating and mechanical testing	Cleaned and stored
W-Cu-03	29.92	9.99	36.89	For coating and mechanical testing	Cleaned and stored
W-Cu-04	30.11	10.00	37.08	Uncoated reference for testing	Cleaned and stored

Characterization Specimens (5 mm)					
W-Cu-05	5.12	10.01	6.22	For coating and microstructural analysis	Cleaned and stored
W-Cu-06	4.95	9.99	6.17	For coating and microstructural analysis	Cleaned and stored
W-Cu-07	5.08	10.00	6.19	For coating and microstructural analysis	Cleaned and stored
W-Cu-08	4.98	10.02	6.18	For coating and microstructural analysis	Cleaned and stored
W-Cu-09	5.05	9.98	6.14	For coating and microstructural analysis	Cleaned and stored
W-Cu-10	4.88	10.00	6.16	For coating and microstructural analysis	Cleaned and stored
W-Cu-11	5.03	10.01	6.20	Uncoated substrate characterization	Cleaned and stored
W-Cu-12	4.92	9.99	6.15	Uncoated substrate	Cleaned

				characterization	and stored
Corrosion Test Specimens (2 mm)					
W-Cu-13	2.05	9.99	2.48	For coating and corrosion testing	Cleaned and stored
W-Cu-14	1.98	10.00	2.45	For coating and corrosion testing	Cleaned and stored
W-Cu-15	2.10	9.98	2.46	For coating and corrosion testing	Cleaned and stored
W-Cu-16	1.95	9.99	2.44	Uncoated reference for corrosion testing	Cleaned and stored

CHAPTER 8: CONCLUSION

The initial phase of the research project on the development and evaluation of a surface-engineered pantograph contact strip has been successfully concluded, establishing a solid and rigorous foundation for all subsequent investigative work. This phase was dedicated to the critical tasks of material selection, methodological design, and precise specimen preparation, all of which are paramount to the validity and success of the overall research aim.

The process commenced with the comprehensive selection and procurement of a certified tungsten-copper (80/20) alloy rod, chosen for its optimal balance of electrical conductivity and mechanical strength, making it an ideal substrate for the proposed composite system. Following this, a strategic specimen design was implemented, resulting in three distinct geometries tailored for specific mechanical, microstructural, and corrosion testing requirements. The precision fabrication of these specimens was executed using Wire Electrical Discharge Machining (Wire EDM), a process selected for its ability to machine the hard and brittle W-Cu composite without inducing mechanical stress, micro-cracks, or thermal damage.

The outcome of this machining process was a full set of sixteen specimens that underwent a stringent quality assessment protocol. This verification confirmed that all specimens adhered strictly to the specified dimensional tolerances and were entirely free from surface or internal defects, thus ensuring their suitability for the critical coating application and performance evaluation stages that will follow.

The meticulous execution of this preparatory phase—encompassing certified material procurement, thoughtful specimen design, precision machining, and rigorous quality control—has resulted in a reliable, well-documented, and reproducible experimental platform. The specimens are now in a state of readiness, having been cleaned, catalogued, and stored under controlled conditions to prevent contamination or degradation. This successful completion of the foundational work not only validates the chosen methodologies but also sets a high standard of procedural rigor. It provides a robust and trustworthy basis upon which the next phase of the project—the application of the HVOF-sprayed Stellite 6 coating and its comprehensive performance characterization—can confidently proceed. The achievement of this milestone marks a significant step towards the ultimate goal of developing a next-generation pantograph contact strip capable of meeting the severe demands of high-speed rail operations.

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